Escape the Geese – Risk Assessment

Cohort 2 Team 3 – ‘Kostas Barmpis and Dimitris Kolovos’

Team members:

Sruthiksha Ananth

Louis Price

Sarah James

Ben Smejko

Henrik Bowden

Tim Giles

June Njiru

Introduction to risk management

Risk assessment was incorporated in the development of Escape the Geese so we could identify and address potential issues that could affect the progress of the project or the quality of the deliverables. To manage the risks, our approach was to first identify potential risks, then analyse their likelihood and severity before planning how to reduce them. Finally, we monitored these throughout the project.

For risk identification, we considered issues that could affect different parts of the project including the team members, technical development, project planning and the game itself. These included team availability, merge conflicts, loss of code, missed deadlines and performance issues. Considering the risks across different categories helped us identify the threats beyond the technical implications of the game.

In risk analysis, each risk was assessed based on how likely it was to happen, and how seriously it might affect the project. We used three ratings for both likelihood and severity, which were either low(L), medium(M) or high(H). For likelihood, Low means unlikely to happen, Medium being plausible and High being very likely to happen. For severity, Low rating for issues that might cause minor delays or inconvenience with little to no effect on the deliverables, medium rating for issues causing noticeable delays or deterioration of the quality of deliverables and high rating for risks causing serious delays or failure to meet the requirements to the project. These ratings helped us identify which parts of the project needed more attention.

Similarly, for risk planning, we identified the measures that could help prevent risks from happening or reduce the impact that it had on the project. For instance using GitHub to store and manage our code to reduce the chances of losing our work. Also, having more than one person familiar with each task so that the progress of the project would not be solely dependent on one team member being available.

Finally, for the risk monitoring, we assigned each risk to a team member to keep track of it and make sure that the planned mitigation measures are followed. We also planned to review the risks during weekly team meetings especially if there are changes or any issues come up, which would give room to update the likelihood and severity ratings where necessary and adjust mitigation measures accordingly.

We chose to organise our risks in a risk register, using a table with the risk ID, category, description, likelihood, severity, mitigation measures and owner, making the information easy to follow and helps us keep track of who is responsible for each risk. Since Escape the Geese is a project with a short development period, we decided that a simple risk register with Low, Medium and High ratings would be sufficient, rather than using a more complicated risk assessment method.

Risk Index

T_ID	Category	Description	Likelihood	Severity	Mitigation	Owner
R_1	Personnel	Family bereavement forces a team member to take a leave of absence causing them to not complete assigned work	M	M	Increase the “bus factor”: any task assigned to a member of the group is always	Louis

					assigned to at least one other member	
R_2	Personnel	Outbreak of illness causes one or more team members to call off sick	M	M	1. Increase the “bus factor”: any task assigned to a member of the group is always assigned to at least one other member. 2.Ill members should attend group meetings virtually to prevent spread of disease	Louis
R_3	personnel	Group member does not complete their assigned task	H	L	Help increase the bus factor by having more than one person responsible for each task. This means any non-completion by a team member can be compensated by their task team leader	Louis
R_4	Technical	Losing code (file corrupts ect)	M	H	Commit to Git often and use github so that in the event of loss of one person's device, the code has multiple backups	Henrik
R_5	Technical	Merge conflicts when working in a team	H	M	Use branches and small commits and agree on who edits what, (use of modularity)	Tim
R_6	Technical	Work with an unfamiliar library	H	L	Prototype early, read documentation and practise	Henrik
R_7	Technical	JAR file causes users browser to crash	M	M	Incorporate heavy testing into the development process, any bottlenecks should be optimised to prevent crashing	Tim
R_8	Planning	Missing Deadline on 16th November	L	H	Have submission “chain of command” to prevent internet issues/ forgetting the deadline	Sruthi
R_9	Planning	Under estimating time for key features and having to rush parts of the implementation	M	H	Plan pessimistically and make sure time is left over (plan for murphy's law)	Sruthi

R_10	Product	Slow startup/ low framerate due to geese	M	M	Cap goose count, incorporate heavy testing to help mitigate slow start issues	June
R_11	Product	Third party assets that are being used are unsuitable for commercial use	L	H	Keep a log of assets and use only assets which have permission for commercial use	June

Key

M =Medium Likelihood/Severity

H = High Likelihood/Severity

L = Low Likelihood/Severity